

SQL Fundamentals

10 Interview Questions — Northwind Database

The Complete DSA + SQL Interview Preparation Series

Book 4 of 13

For placement preparation, coding interviews, and product-based company rounds.

About This Book

This book opens the SQL section of the series, built entirely on the Northwind database — Customers, Categories, Employees, OrderDetails, Orders, Products, Shippers, and Suppliers. This is the schema most commonly used in SQL interview rounds because it mirrors a real order-management system: customers place orders, orders contain line items, line items reference products, products belong to categories and suppliers.

Each question here starts with a business scenario the way a hiring manager or interviewer would actually phrase it, followed by the tables you'll need and the columns your query is expected to return. No solutions are provided — writing the query is the exercise.

How To Use This Book

Read the scenario and picture the person asking for it — a sales manager, an ops lead, a finance analyst. Identify which tables are involved and how they join before writing a single line of SQL. Check your result's columns against 'Expected Output' to make sure you're answering exactly what was asked, not a nearby question.

Difficulty Breakdown

Difficulty	Count
Easy	8
Medium	2

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Q1. List All Customers

Easy

Topic: SELECT / Basics

The support team wants a full contact list of every customer in the system, showing company name, contact name, and country, sorted alphabetically by company name.

TABLES USED

Customers

EXPECTED OUTPUT

CompanyName, ContactName, Country

Q2. Products Currently Out of Stock

Easy

Topic: WHERE / Filtering

The warehouse team needs a quick list of every product whose current stock (UnitsInStock) is zero, so they can prioritize reordering.

TABLES USED

Products

EXPECTED OUTPUT

ProductName, UnitsInStock

Q3. Employees Hired After a Given Date

Easy

Topic: WHERE / Date Filtering

HR wants to identify all employees hired after January 1, 1994, for a tenure-based bonus review.

TABLES USED

Employees

EXPECTED OUTPUT

FirstName, LastName, HireDate

Q4. Count of Products per Category

Easy

Topic: GROUP BY / COUNT

The merchandising team wants to know how many distinct products fall under each product category, to spot categories that are under-stocked with variety.

TABLES USED

Products, Categories

EXPECTED OUTPUT

CategoryName, ProductCount

Q5. Top 5 Most Expensive Products

Easy

Topic: ORDER BY / LIMIT

The pricing team wants the five most expensive products currently listed, to double-check they're priced correctly before a catalog refresh.

TABLES USED

Products

EXPECTED OUTPUT

ProductName, UnitPrice

Q6. Customers From a Specific Country

Easy

Topic: WHERE / Filtering

A regional sales rep needs the list of all customers based in Germany, including their city, to plan an in-person visit schedule.

TABLES USED

Customers

EXPECTED OUTPUT

CompanyName, City, Country

Q7. Orders Placed by a Specific Customer

Easy

Topic: JOIN / Basics

A customer success manager wants every order placed by a specific customer, showing when it was ordered and whether it has shipped yet.

TABLES USED

Orders, Customers

EXPECTED OUTPUT

OrderID, OrderDate, ShippedDate

Q8. Suppliers Without a Fax Number

Easy

Topic: WHERE / NULL Handling

The procurement team is cleaning up supplier records and wants a list of every supplier that has no fax number on file, since that field is now used for automated order confirmations.

TABLES USED

Suppliers

EXPECTED OUTPUT

CompanyName, ContactName, Phone

Q9. Average Order Value per Customer

Medium

Topic: JOIN / GROUP BY / AVG

The finance team wants to understand customer value and asks for the average order value (based on unit price times quantity across all line items) for each customer, sorted from highest to lowest.

TABLES USED

Customers, Orders, OrderDetails

EXPECTED OUTPUT

CompanyName, AverageOrderValue

Q10. Products Never Ordered

Medium

Topic: LEFT JOIN / NULL Check

The merchandising team wants to identify dead stock — products that have never appeared in a single order — so they can consider discontinuing them.

TABLES USED

Products, OrderDetails

EXPECTED OUTPUT

ProductName, UnitsInStock